

Computer Graphics Term Project

Important Notes

- **Deadline week starting from Wednesday 13/5 till Thursday 14/5**
- **Submission will be in your Lab**
- **No late submissions**
- **-10 for copies**
- **Groups from 3 to 5 students**
- **Be ready to be asked about each function**
- For any question, please contact directly to TA in your lab time

Requirements [10 marks]:

Implement 2D drawing package which uses menus/drop down list to provide the following features:

1) File Menu:

- a) Implement function to clear screen from shapes
- b) Implement function to save all data drawn on the screen to a file
- c) Implement function to load saved data from files

2) Preferences Menu:

- a) Change the background of window to be white
- b) Change the shape of your window mouse
- c) Give the user option to choose shape color before drawing from menu

3) Lines Menu:

- a) Implement line algorithms [DDA, Midpoint and Parametric]

4) Circles Menu:

- a) Implement Circle algorithms (Direct, Polar, iterative Polar, midpoint and modified Midpoint)

5) Ellipse Menu:

- a) Ellipse Algorithms [Direct, polar and midpoint]

6) Curves Menu:

- a) Cardinal Spline Curve

7) Filling Menu:

- a) Filling Circle with lines after taking filling quarter from user
- b) Filling Circle with other circles after taking filling quarter from user
- c) Filling Square with Hermit Curve[Vertical]
- d) Filling Rectangle with Bezier Curve[horizontal]
- e) Convex and Non-Convex Filling Algorithm
- f) Recursive and Non-Recursive Flood Fill

8) Clipping Menu:

- a) Clipping algorithm using Rectangle as Clipping Window [Point ,Line, Polygon]
 - b) Clipping algorithm using Square as Clipping Window [Point ,Line]
- 9) User must interact with the window using mouse only to draw the shapes
- 10) Use the console to display logs/information + take an input option from the user

Bonus: [4 marks]

- 1. Clipping algorithm using circle as a Clipping Window [Point, Line]
- 2. Draw **smiley faces** using **circles as the face and eyes, curves as the mouth** and **lines for the nose**:
 - a. Happy Face
 - b. Sad Face